

DSPO: STABLE AND EFFICIENT POLICY OPTIMIZATION FOR AGENTIC SEARCH AND REASONING

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ABSTRACT

External search enables LLMs to answer questions whose evidence lies beyond static parametric memory, but training such agents by reinforcement learning remains unstable because sparse trajectory-level rewards are typically optimized with token-level objectives. We introduce **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, an RL algorithm for search-augmented LLM agents that combines a sequence-level importance ratio with dynamic outcome-based filtering of rollout groups. The sequence-level update aligns policy correction with terminal rewards, while the filter retains only groups containing both successful and unsuccessful trajectories, preserving a non-degenerate advantage signal under sparse binary feedback. Without supervised demonstration trajectories, DSPO learns to interleave reasoning and multi-turn search. Across seven open-domain QA benchmarks with a BM25 retriever, the DSPO-trained 7B model improves over the comparable Search-R1 7B GRPO baseline by **34.1%** on average and, on HotpotQA, exceeds the Search-R1 14B GRPO model by nearly **9% relative**, while the training curves show substantially more stable optimization than token-level alternatives.

1 INTRODUCTION

Large Language Models (LLMs) (Brown et al., 2020; Touvron et al., 2023; Zhao et al., 2023) have become highly capable across domains such as mathematics (Shao et al., 2024; Trinh et al., 2024; Yu et al., 2025), coding (Zheng et al., 2023; Yang et al., 2024), and creative writing (Chakrabarty et al., 2024; Marco et al., 2024). Nevertheless, their knowledge remains bounded by the data encoded during pretraining and subsequent adaptation. A direct way to relax this constraint is to equip LLMs with external search, thereby converting generation into an interactive process in which the model can acquire evidence at inference time (Jin et al., 2025b). This setting is a concrete form of tool calling (Schick et al., 2023): the model must decide when to continue internal reasoning, when to issue a search query, and how to integrate retrieved evidence into the final answer. Because these decisions unfold over a history of reasoning and tool interactions, agentic search is naturally formulated as a sequential decision-making problem for Reinforcement Learning (RL).

RL is particularly attractive in this setting because, unlike Supervised Fine-Tuning (SFT), it does not require a fixed corpus of expert search demonstrations and can improve policies through trial-and-error interaction (Chu et al., 2025). Value-free methods such as Group Relative Policy Optimization (GRPO) (Shao et al., 2024) further reduce memory overhead by replacing learned value models with reward statistics computed over sampled response groups. However, interactive search also exposes two optimization failures that are consequential for LLM agents (Jin et al., 2025b; Yu et al., 2025; Cui et al., 2025; Liu et al., 2025). First, as Zheng et al. (2025) note, token-level importance ratios are poorly aligned with rewards assigned to complete responses or trajectories, yielding high-variance updates that can precipitate policy collapse. Second, binary terminal rewards often produce homogeneous rollout groups in which every trajectory succeeds or every trajectory fails; after group-relative normalization, such groups provide little or no usable advantage signal (Yu et al., 2025; Liu et al., 2025).

We address these coupled sources of instability and sample inefficiency with **Dynamic-filter Sequence-level Policy Optimization (DSPO)**. The contribution is deliberately focused: DSPO adapts and combines sequence-level policy correction and outcome-based filtering into a single RL objective for agentic search. It uses a GSPO-style sequence-level ratio (Zheng et al., 2025) so that the policy update is computed at the same granularity as the terminal reward, and it uses a DAPO-inspired filter (Yu et al., 2025) to retain rollout groups only when they contain both successful and unsuccessful trajectories. This combination preserves informative advantages while avoiding the token-level ratio fluctuations that destabilize long search trajectories. Under the evaluation protocol used in this paper, the resulting 7B model achieves a **34.1% relative improvement** over the comparable 7B Search-R1 GRPO baseline (Jin et al., 2025b).

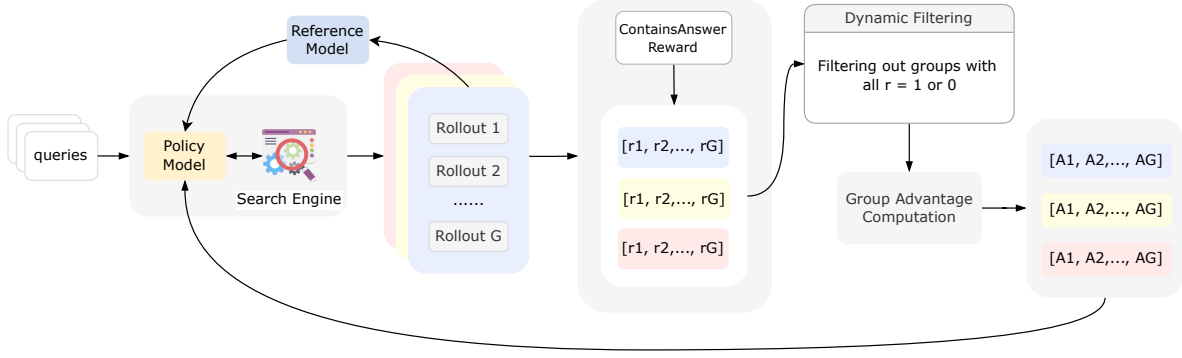


Figure 1: Overview of the DSPO training loop. For each query, the policy generates a group of G search-and-reasoning trajectories through interaction with the environment, and each trajectory receives a sparse terminal reward. The **dynamic filter** discards groups with homogeneous outcomes and continues sampling until the batch is filled, so every retained group provides a non-degenerate group-relative advantage. The policy is then updated with the sequence-level objective.

and exceeds the corresponding 14B GRPO system (Jin et al., 2025a) on HotpotQA by **nearly 9% relative** (0.613 vs. 0.563).

The paper makes three contributions that connect the optimization mechanism to its empirical consequences:

- We introduce DSPO, an RL algorithm for agentic search that combines **sequence-level optimization** with **dynamic outcome-based filtering** to address token-level update instability and sparse-reward sample inefficiency in a single training procedure.
- We evaluate DSPO on seven QA benchmarks and show that the 7B model delivers a **34.1% relative improvement** over the comparable 7B Search-R1 GRPO baseline and exceeds the 14B GRPO counterpart on HotpotQA, with a nearly **9% relative gain** (0.613 vs. 0.563).
- We provide evidence that the gains are attributable to the optimization design rather than retrieval engineering: all RL experiments use a standard BM25 retriever, and ablations isolate the respective effects of sequence-level correction and dynamic filtering.

2 RELATED WORK

2.1 RL FOR LLMs

RL for LLMs has evolved from Reinforcement Learning from Human Feedback (RLHF) methods based on PPO and explicit reward models (Ouyang et al., 2022; Christiano et al., 2017; Schulman et al., 2017) toward simpler direct-optimization frameworks such as DPO (Rafailov et al., 2023). Group Relative Policy Optimization (GRPO) (Shao et al., 2024) is an important value-free alternative because it replaces a learned value function with reward statistics over sampled response groups, but its token-level objective can induce instability (Liu et al., 2025; Cui et al., 2025). Recent work addresses complementary limitations of this paradigm: GSPO moves the correction ratio to the sequence level so that optimization better matches the reward unit (Zheng et al., 2025), DAPO improves sparse-reward learning through dynamic outcome-based sampling (Yu et al., 2025), and GMPO uses geometric-mean aggregation to reduce outlier sensitivity in token-level optimization (Zhao et al., 2025). DSPO is positioned as an integration of these relevant principles for autonomous search: it combines sequence-level policy correction with dynamic filtering so that each update is both trajectory-aligned and supported by a non-degenerate reward comparison.

2.2 LLMs WITH AGENTIC RETRIEVAL

Retrieval-augmented generation (RAG) mitigates the static knowledge limitations of LLMs by conditioning generation on information retrieved from external corpora (Lewis et al., 2020; Gao et al., 2023). Classical RAG pipelines typically retrieve documents, often with dense retrievers, and concatenate them with the input context (Karpukhin et al., 2020); this design supplies evidence but usually fixes the retrieval policy. Agentic retrieval instead lets the model plan,

search, and reason over multiple turns. ReAct couples reasoning traces with tool invocation (Yao et al., 2023), and multi-agent systems such as AutoGen support collaborative tool-based workflows (Wu et al., 2024). More recent agentic RAG and RL methods learn search behavior through decision making: Wu et al. (2025) propose Agentic Reasoning with external tools, while RL-based approaches (Jin et al., 2025b; Chen et al., 2025; Song et al., 2025) train LLMs to interleave search and reasoning without supervised demonstrations. This end-to-end setting reduces reliance on hand-designed retrieval schedules, but it makes optimization stability and sparse terminal rewards central technical constraints. DSPO targets these constraints within the same agentic-search regime.

3 METHODOLOGY

This section formalizes agentic search as an RL problem, reviews the policy-optimization objectives on which DSPO builds, and identifies why token-level correction is poorly matched to sparse trajectory-level rewards. We then define **Dynamic-filter Sequence-level Policy Optimization (DSPO)**, specifying how sequence-level correction and dynamic outcome filtering jointly support stable, sample-efficient policy updates before presenting the complete training algorithm.

3.1 PRELIMINARIES

Policy Gradient Methods for LLMs. RL training for LLMs commonly relies on policy-gradient methods such as PPO (Schulman et al., 2017), which is widely used for alignment. PPO updates a policy π_θ by maximizing a clipped surrogate objective on samples generated by an earlier policy $\pi_{\theta_{\text{old}}}$. In the token-averaged form considered here, the objective is:

$$J_{\text{PPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{|y|} \sum_{t=1}^{|y|} \min \left(r_t(\theta) \hat{A}_t, \text{clip} \left(r_t(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_t \right) \right], \quad (1)$$

where $r_t(\theta) = \frac{\pi_\theta(y_t|x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t|x, y_{<t})}$ denotes the token-level importance ratio. PPO also requires a separately trained value model to estimate token-level advantages $\hat{A}_t$ through Generalized Advantage Estimation (GAE) (Schulman et al., 2015), increasing memory cost and introducing another potential source of instability.

GRPO (Shao et al., 2024) removes the value model by sampling a group of G responses $\{y_i\}_{i=1}^G$ for each prompt x and computing each response advantage from reward statistics within the group. Although this design substantially simplifies training, the standard GRPO objective still applies policy correction at token granularity:

$$J_{\text{GRPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, \{y_i\}_{i=1}^G \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \min \left(r_{i,t}(\theta) \hat{A}_i, \text{clip} \left(r_{i,t}(\theta), 1 - \epsilon, 1 + \epsilon \right) \hat{A}_i \right) - \beta D_{KL}(\pi_\theta || \pi_{\text{ref}}) \right], \quad (2)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})}$, and the advantage for every token $y_{i,t}$ in a response y_i is set to the same sequence-level value:

$$\hat{A}_{i,t} = \hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R)}, \quad (3)$$

Thus, every token in response y_i receives the same advantage $\hat{A}_i$, although that advantage is computed from a sequence-level reward. The resulting mismatch between token-level ratios and sequence-level feedback is the central optimization issue addressed by DSPO.

Agentic Search as a Markov Decision Process. We model agentic search and reasoning as a discrete-time, finite-horizon Markov Decision Process (MDP), in which each decision is conditioned on the accumulated interaction history.

- **State (s_t):** At turn t , the state s_t encodes the entire interaction history: the initial question q , all previously generated thoughts and search queries, and the retrieved evidence returned by the environment. Because search results must be interpreted and integrated into subsequent decisions, the state necessarily grows with the trajectory, creating long-horizon dependencies that conventional token-level RL struggles to optimize.

- **Action (a_t):** An action is a full textual segment generated by the policy π_θ , consisting of free-form reasoning followed by a decision. The action terminates either with a `</tool_call>` token, which triggers a search, putting the results within `</tool_response>`, or with a `</answer>` token, which ends the trajectory. This structured action space forces the agent to learn not only *what* to generate but also *when* to search—an aspect that introduces significant variability in trajectory length.
- **Policy (π_θ):** The policy π_θ is the underlying LLM, generating tokens autoregressively conditioned on the state. The policy conditions on the full history, but the optimization target is calculated only on the model-generated thoughts and actions, masking out the retrieved content from the environment (Jin et al., 2025b). Optimizing this policy requires credit assignment over long sequences in which many intermediate reasoning steps do not receive direct supervision or reward, further emphasizing the need for stable sequence-level optimization.
- **Trajectory (τ):** A trajectory $\tau = (s_1, a_1, \dots, s_T, a_T)$ records all reasoning and tool interactions taken for a given question, including both model-generated actions and environment-returned search results. Because the final reward is assigned at the level of the entire trajectory, the optimization problem is fundamentally sequence-level: every early decision can influence the eventual answer correctness.
- **Reward ($R(\tau)$):** We employ a sparse terminal reward: a trajectory receives $R = 1$ if the final answer contains the ground-truth text, and $R = 0$ otherwise:

$$R(\tau) = \begin{cases} 1 & \text{if } a_{\text{gold}} \subseteq a_{\text{pred}}, \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

RL with a Search Engine. Following Jin et al. (2025b), we explicitly model the search engine, denoted as $\mathcal{S}$, as part of the environment. The policy LLM π_θ learns to generate trajectories by interleaving reasoning with calls to $\mathcal{S}$. The overall optimization problem is to find a policy that maximizes the expected reward, regularized by a KL divergence term to prevent large deviations from a reference policy π_{ref} :

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x; \mathcal{S})} [R(x, y)] - \beta D_{\text{KL}} [\pi_\theta(y|x; \mathcal{S}) || \pi_{\text{ref}}(y|x; \mathcal{S})]. \quad (5)$$

Here, $y \sim \pi_\theta(\cdot|x; \mathcal{S})$ signifies that the trajectory y is generated through a multi-step process involving both the policy’s token generation and the information returned by the search engine $\mathcal{S}$.

Motivation for DSPO. While frameworks like Search-R1 (Jin et al., 2025b) have successfully framed agentic search as an RL problem, applying conventional algorithms like PPO or GRPO faces significant hurdles. The open-ended nature of the search environment exacerbates the instability of token-level optimization. A core issue is the fundamental mismatch between the unit of sequence-level reward assignment and the unit of token-level optimization (Zheng et al., 2025). This discrepancy leads to high-variance gradient estimates that accumulate over long trajectories, often culminating in policy collapse. Furthermore, the sparse binary reward signal means many training batches may contain only successful or only unsuccessful trajectories, yielding abnormal advantage and thus providing no learning signal, which drastically reduces sample efficiency (Yu et al., 2025; Liu et al., 2025). DSPO is designed to directly counteract these two critical failure modes.

3.2 DYNAMIC-FILTER SEQUENCE-LEVEL POLICY OPTIMIZATION

DSPO introduces two key innovations over prior methods: (1) it performs policy optimization at the sequence level, aligning the training objective with the trajectory-based reward structure, and (2) it incorporates a dynamic filtering mechanism to ensure every training batch provides a high-quality, non-zero learning signal. The entire training process, which integrates these components, is depicted in Figure 1.

3.2.1 SEQUENCE-LEVEL POLICY OPTIMIZATION FOR ENHANCED STABILITY

Inspired by GSPO (Zheng et al., 2025), we replace the unstable token-level importance ratio with a theoretically grounded sequence-level counterpart. The sequence-level importance ratio $s_i(\theta)$ for a response y_i is defined as the geometric mean of its token-level ratios:

$$s_i(\theta) = \left(\frac{\pi_\theta(y_i|x)}{\pi_{\theta_{\text{old}}}(y_i|x)} \right)^{\frac{1}{|y_i|}} = \exp \left(\frac{1}{|y_i|} \sum_{t=1}^{|y_i|} \log \frac{\pi_\theta(y_{i,t}|x, y_{i,<t})}{\pi_{\theta_{\text{old}}}(y_{i,t}|x, y_{i,<t})} \right). \quad (6)$$

This length normalization is crucial for reducing variance and ensuring that $s_i(\theta)$ remains within a consistent numerical range regardless of sequence length, which is vital for stable clipping.

Gradient Analysis. The gradient analysis below shows why DSPO enhances the stability. The gradient of the token-level GRPO objective (unclipped) scales each token’s log-probability gradient by a noisy, token-specific weight $r_{i,t}(\theta)$. In contrast, the gradient of our sequence-level objective scales the average log-probability gradient of the entire sequence by a single, more stable sequence-level weight $s_i(\theta)$:

$$\nabla_{\theta} J_{\text{GRPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot \sum_{t=1}^{|y_i|} r_{i,t}(\theta) \nabla_{\theta} \log \pi_{\theta}(y_{i,t} | \dots) \right] \quad (7)$$

$$\nabla_{\theta} J_{\text{DSPO}} \propto \mathbb{E} \left[\hat{A}_i \cdot s_i(\theta) \cdot \sum_{t=1}^{|y_i|} \nabla_{\theta} \log \pi_{\theta}(y_{i,t} | \dots) \right] \quad (8)$$

By applying a single, holistic correction factor to the entire trajectory, DSPO avoids the accumulation of token-level noise that plagues prior methods, leading to fundamentally more stable training. In parallel, the dynamic filtering mechanism guarantees a normal advantage signal $\hat{A}_i$ by constructing training batches from rollout groups that contain both successes and failures, thus preventing wasted samples in sparse-reward environments.

3.2.2 DYNAMIC OUTCOME-BASED FILTERING FOR EFFICIENT LEARNING

The sparse binary nature of our reward function poses a challenge for group-based advantage estimation. If all G responses in a group are correct ($R = 1$) or all are incorrect ($R = 0$), the normalized advantage $\hat{A}_i$ becomes zero or undefined. Such batches do not provide a useful gradient signal, wasting computational resources.

To overcome this, DSPO incorporates a dynamic filtering mechanism inspired by DAPO (Yu et al., 2025). During sampling, we only retain groups of trajectories that contain a mix of successful and unsuccessful outcomes. A group $\{y_i\}_{i=1}^G$ is used for training only if its rewards $\{R_i\}_{i=1}^G$ satisfy:

$$0 < \sum_{i=1}^G R_i < G. \quad (9)$$

This ensures that the reward variance within every training group is non-zero, guaranteeing a meaningful advantage signal. This dynamic selection curates a high-quality dataset for each policy update, transforming a sparse reward problem into a dense and efficient learning signal.

3.3 THE DSPO OBJECTIVE AND TRAINING ALGORITHM

By integrating these components, we arrive at the final DSPO objective. For each valid group from the filtered sample space $\mathcal{D}_{\text{filtered}}$, we compute the advantage $\hat{A}_i$ using group-relative normalization:

$$\hat{A}_i = \frac{R_i - \text{mean}(R)}{\text{std}(R) + \delta}, \quad (10)$$

where δ is a small constant for numerical stability. The policy π_{θ} is updated by maximizing:

$$J_{\text{DSPO}}(\theta) = \mathbb{E}_{(x, \{y_i\}) \in \mathcal{D}_{\text{filtered}}} \left[\frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i \right) \right], \quad (11)$$

where s_i is the sequence-level importance ratio defined as the geometric mean of the token-level ratios. Crucially, during likelihood calculation, we apply loss masking to all tokens retrieved from the search tool following Jin et al. (2025b). This ensures the model learns to utilize external knowledge for reasoning, not simply to reproduce it. The full training process is detailed in Algorithm 1.

4 EXPERIMENTS

In this section, we conduct a series of experiments to empirically validate the effectiveness of our proposed Dynamic-filter Sequence-level Policy Optimization (DSPO) algorithm. Our primary objectives are to demonstrate that DSPO: (1) achieves exceptional performance on challenging question-answering benchmarks; (2) exhibits significantly enhanced training stability, avoiding the catastrophic collapse that plagues baseline methods; and (3) derives its performance gains from the synergistic combination of its core components.

Algorithm 1 Dynamic-filter Sequence-level Policy Optimization (DSPO)

```
1: Input: Initial policy  $\pi_{\theta_0}$ , fixed reference policy  $\pi_{\text{ref}}$ , prompt dataset  $\mathcal{D}$ , group size  $G$ , batch size  $B$ , search tool  $\mathcal{R}$ .
2: Initialize policy  $\pi_{\theta} \leftarrow \pi_{\theta_0}$ .
3: for each training step do
4:    $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$ .
5:   Initialize training buffer  $\mathcal{B} \leftarrow \emptyset$ .
6:   while  $|\mathcal{B}| < B$  do
7:     Sample a prompt  $x \sim \mathcal{D}$ .
8:     Generate a group of  $G$  trajectories  $\{y_i\}_{i=1}^G$  using  $\pi_{\theta_{\text{old}}}$  and the search tool  $\mathcal{R}$ .
9:     Compute terminal rewards  $\{R_i\}_{i=1}^G = \{\text{ContainsAnswer}(y_i, y_{\text{gold}})\}_{i=1}^G$ .
10:    if  $0 < \sum_{i=1}^G R_i < G$  then ▷ Dynamic outcome-based filtering
11:      Add  $(x, \{y_i\}_{i=1}^G, \{R_i\}_{i=1}^G)$  to  $\mathcal{B}$ .
12:    end if
13:  end while
14:  for each  $(x, \{y_i\}, \{R_i\})$  in  $\mathcal{B}$  do
15:    Compute advantages  $\{\hat{A}_i\}_{i=1}^G$  via group normalization of  $\{R_i\}$ .
16:    Compute sequence-level importance ratios  $\{s_i(\theta)\}_{i=1}^G$  using Eq. 6.
17:    Compute the DSPO loss for the group using Eq. 11, applying masks to retrieved tokens.
18:  end for
19:  Update policy parameters  $\theta$  by taking a gradient step on the total loss from  $\mathcal{B}$ .
20: end for
```



Figure 2: **Validation performance of DSPO across seven benchmarks during training.** The steady, monotonic increase in accuracy confirms that DSPO’s reward improvement translates directly to enhanced generalization and that our method learns a robust search-and-reasoning policy.

4.1 EXPERIMENTAL SETUP

Following Search-R1 (Jin et al., 2025b), As shown in Table 1, we use the prompt template to instruct the model’s actions during the search task, including `<think>`, `<tool_call>` and `<answer>`.

Benchmarks and Baselines. To provide a rigorous evaluation, our experimental design adheres to the established protocol of Search-R1 (Jin et al., 2025b). We train our model on a composite dataset containing the training splits of Natural Questions (NQ) (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018). We then assess its generalization capabilities on the test sets of seven diverse QA benchmarks: NQ, TriviaQA (Joshi et al., 2017), PopQA (Mallen et al., 2022), HotpotQA, 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022), and Bamboogle (Press et al., 2022).

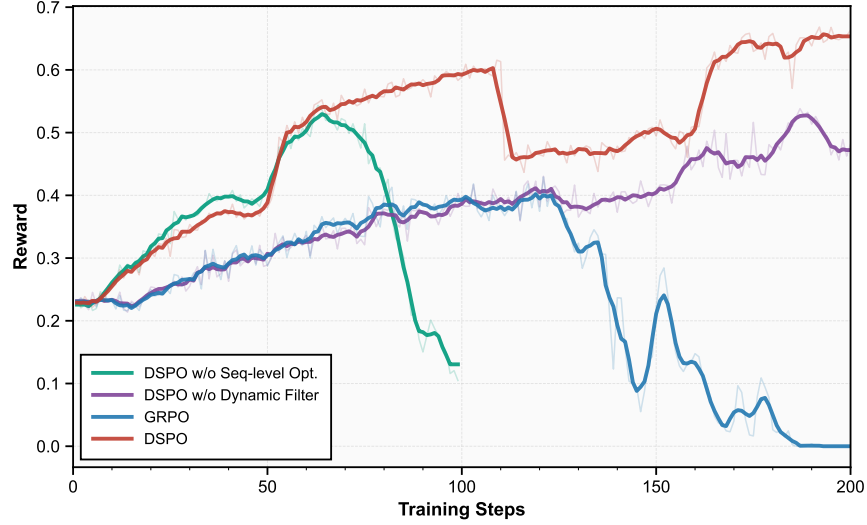


Figure 3: **Training reward dynamics of DSPO and its ablations.** Comparative view of learning curves. DSPO (red) demonstrates stable and monotonic improvement. In contrast, token-level variants (green, blue) suffer catastrophic policy collapse, while the sequence-level variant without our filter (purple) plateaus at a suboptimal level.

Prompt Template. Answer the given question. You must conduct reasoning inside `<think>` and `</think>`. first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<tool_call>` query `</tool_call>` and it will return the top searched results between `<tool_response>` and `</tool_response>`. You can search as many times as your want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>`, without detailed illustrations. For example, `<answer>` Beijing `</answer>`. **Question:** ...

Table 1: The prompt template used in our experiments.

Our comparison suite includes strong external baselines and critical internal ablations. External baselines are the Qwen2.5-7B and 14B models trained with PPO and GRPO from the Search-R1 framework (Jin et al., 2025b;a). To deconstruct our method, we also include two internal baselines as ablations: (1) **DSPO w/o dynamic filter**, which is equivalent to GSPO (Zheng et al., 2025), and (2) **DSPO w/o sequence-level opt.**, which reverts to a strong token-level policy, DAPO (Yu et al., 2025).

For implementation, we used Qwen2.5-7B-Instruct model as the starting checkpoint for all our training. Our experiments are built upon the VeRL framework (Sheng et al., 2025), for which we adapted the provided Search-R1-like example code and scripts to suit our methodology. We benchmark DSPO against a comprehensive suite of baselines. For external comparison, we use the PPO and GRPO methods from the Search-R1 framework (Jin et al., 2025b;a). Crucially, as our work utilizes a modified reward function, we retrained these models under our exact experimental conditions to ensure a fair comparison. The results of these retrained models serve as our primary external benchmarks.

Implementation and Evaluation. To isolate the benefits of our algorithm, all RL experiments deliberately employ a standard BM25 retriever. This controlled setup ensures that observed performance improvements are directly attributable to the model’s learned policy. Across all methods, models are trained using a sparse, binary reward signal based on substring Exact Match (subEM) of the final answer, and subEM serves as the primary evaluation metric.

4.2 MAIN RESULTS AND ABLATION STUDY

To provide a holistic view of our algorithm’s effectiveness, we present a comprehensive comparison in Table 2. Due to the synergistic nature of DSPO’s components, we find it most illustrative to present our main results alongside our ablation study. This single table juxtaposes DSPO against both external state-of-the-art baselines and its own ablated variants, offering a clear and direct assessment of its overall superiority and the indispensability of its core components.

Table 2: Comprehensive comparison of DSPO with baselines and ablation variants on seven QA benchmarks. Baselines include Search-R1 models (7B & 14B) trained with GRPO and PPO (Jin et al., 2025b;a). Ablations remove key components: ‘w/o dynamic filter’ and ‘w/o seq-level opt.’. Original EM scores from Search-R1 are in parentheses. To maintain the consistency of evaluation, we retrained and evaluated them using our adjusted rewards. Best results are in **bold**; second-best are underlined.

Dataset	Search-R1				DSPO & Ablations (Ours, 7B)		
	GRPO (7B)	PPO (7B)	GRPO (14B)	PPO (14B)	w/o dyn. filter	w/o seq-lvl opt.	DSPO
NQ	0.423 (0.429)	(0.393)	<u>0.535</u> (0.482)	(0.424)	0.363	0.470	0.580
TriviaQA	0.658 (0.623)	(0.610)	0.760 (0.667)	(0.660)	0.515	0.695	<u>0.754</u>
PopQA	0.395 (0.427)	(0.397)	<u>0.477</u> (0.434)	(0.442)	0.277	0.430	0.498
HotpotQA	0.401 (0.386)	(0.370)	<u>0.563</u> (0.429)	(0.436)	0.330	0.438	0.613
2WikiMultiHopQA	0.357 (0.414)	(0.346)	0.611 (0.424)	(0.379)	0.285	0.398	<u>0.569</u>
Musique	0.122 (0.162)	(0.146)	<u>0.260</u> (0.191)	(0.210)	0.105	0.133	0.270
Bamboogle	0.280 (0.400)	(0.368)	0.504 (0.492)	(0.480)	0.288	0.280	<u>0.432</u>
Average	0.377 (0.396)	(0.385)	<u>0.530</u> (0.446)	(0.433)	0.313	0.406	0.531

Comparison with Baselines. The results in Table 2 underscore DSPO’s clear superiority. Our DSPO-trained 7B agent achieves a remarkable average score of 0.531, establishing a new state-of-the-art. This represents a 34.1% relative improvement over the same-sized Search-R1 (GRPO, 7B) model. More strikingly, our 7B agent achieves a slightly better average score than the much larger Search-R1 14B models (both GRPO and PPO). This outcome provides strong evidence that the performance gains stem from a more effective and stable learning algorithm rather than an over-reliance on model scale.

Analysis of Ablations. The ablation results, also presented in Table 2, unequivocally demonstrate that both of DSPO’s components are indispensable. First, removing the dynamic filter (‘w/o dynamic filter’) causes a catastrophic drop in performance, with the average score plummeting to 0.313. This highlights its critical role; without the filter, the sequence-level objective is starved of a useful learning signal due to homogeneous-reward batches. Second, ablating sequence-level optimization (‘w/o sequence-level opt.’) also leads to a significant performance degradation, yielding an average score of 0.406. While this token-level variant outperforms the filter-less one, it falls well short of the full DSPO model. As we show in the next section, it is also prone to catastrophic training instability. This confirms that the synergy is crucial: sequence-level updates are essential for stability, while our dynamic filter is critical for transforming sparse rewards into an efficient learning signal.

Beyond quantitative metrics, we observe that DSPO enables sophisticated search behaviors, including recognize irrelevant results, query reformulation and multi-turn verification (see Appendix A for detailed trajectory examples). All of these behaviors are emerging from pure RL training through DSPO.

4.3 ANALYSIS OF TRAINING DYNAMICS

To empirically validate our claims regarding stability and efficiency, we analyze the training reward dynamics of DSPO, its ablations, and key baselines. Figure 3 offers a compelling visualization of these dynamics, reinforcing our core architectural choices.

DSPO (red) exhibits a smooth, monotonic ascent, efficiently converging to the highest reward level. This trajectory empirically confirms the stability afforded by its sequence-level objective. In stark contrast, the **token-level methods**—DSPO w/o Seq-level Opt. (green) and vanilla GRPO (blue)—suffer from catastrophic policy collapse early in training. Their rewards plummet after a brief initial improvement, a clear manifestation of the instability caused by high-variance, token-level gradient updates. Meanwhile, **DSPO w/o Dynamic Filter** (purple), which leverages sequence-level updates but lacks an efficient learning signal, remains stable but plateaus at a significantly suboptimal performance ceiling. These dynamics reveal that DSPO’s synergy of sequence-level stability and dynamic filtering is key to its robust and effective policy optimization.

To ensure these improvements in training reward translate to genuine generalization rather than reward hacking, we track validation performance on key benchmarks throughout training. As illustrated in Figure 2, DSPO’s validation

accuracy on NQ, HotpotQA, and other diverse benchmarks rises consistently, mirroring its stable reward curve. This correlation confirms that the agent is learning a generalizable search-and-reasoning policy.

4.4 SCALABILITY AND GENERALIZATION ANALYSIS

To further validate the robustness of our approach, we extend our evaluation to explore model scalability and domain generalization.

Scalability to Larger Models. We investigate whether the stability benefits of DSPO translate to larger parameter scales by training Qwen2.5-14B-Instruct. As detailed in Table 3, DSPO demonstrates remarkable scalability. The DSPO-trained 14B model achieves an average accuracy of 60.6%, significantly outperforming the strong GRPO-14B baseline (53.0%) by a relative margin of **14.3%**. These results confirm that our method effectively leverages increased model capacity, establishing an outperforming performance that consistently exceeds standard baselines.

Table 3: Scalability analysis on Qwen2.5-14B-Instruct. Best results are in **bold**.

Dataset	Instruct (14B)	GRPO (14B)	DSPO (7B)	DSPO (14B)	Gain
NQ	0.345	0.535	0.580	0.629	+17.6%
HotpotQA	0.407	0.563	0.613	0.665	+18.1%
2WikiMQA	0.332	0.611	0.569	0.699	+14.4%
Bamboogle	0.328	0.504	0.432	0.544	+7.9%
PopQA	0.364	0.477	0.498	0.545	+14.3%
TriviaQA	0.643	0.760	0.754	0.802	+5.5%
Musique	0.151	0.260	0.270	0.361	+38.8%
Average	0.367	0.530	0.531	0.606	+14.3%

Generalization to Mathematical Reasoning. We further assess the universality of DSPO by applying it to single-turn mathematical reasoning tasks using the Qwen2.5 and Qwen3 model family. Table 4 presents the comparison on Math500 and Olympiad-Bench. DSPO consistently surpasses GRPO across both 7B and 4B model sizes. This indicates that DSPO are effective for general reasoning domains.

Table 4: Generalization to mathematical reasoning. Best results are in **bold**.

Model	Benchmark	Steps	GRPO	DSPO	Gain
Qwen2.5-Math-7B	Math500	200	0.772	0.798	+2.6%
Qwen3-4B	Olympiad-Bench	100	0.728	0.755	+2.7%

5 CONCLUSION

In this work, we tackled the critical instability and sample inefficiency issues that plague RL for autonomous LLM search agents. We introduced Dynamic-filter Sequence-level Policy Optimization (DSPO), an improved algorithm that ensures robust training through two key components: sequence-level optimization to prevent catastrophic policy collapse, and a dynamic outcome-based filter to transform sparse rewards into a consistently effective learning signal. Our experiments demonstrated that DSPO not only achieves substantial performance across a suite of challenging question-answering benchmarks but also exhibits superior training stability compared to prior methods.

By enabling robust training from environmental feedback alone, DSPO establishes a practical and efficient blueprint for creating capable LLM agents without costly expert data. With this stable foundation, future work can confidently explore integrating advanced retrievers or extending DSPO to complex, multi-tool tasks. Furthermore, since the challenges of sparse rewards and unstable policy gradients are not unique to search, we hypothesize that DSPO’s principles will yield similar performance and stability gains in other domains such as mathematics and code generation, which remains a promising direction for future validation. We believe the core tenets of DSPO—matching the optimization unit to the reward signal and guaranteeing signal density—will be instrumental in developing the next generation of autonomous AI.

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A APPENDIX

To provide insight into the learned search strategies, Table A.1 presents representative trajectories from our model. These examples demonstrate the model’s ability to: (1) reformulate queries when initial searches yield irrelevant results, (2) verify information through follow-up searches, and (3) extract comprehensive answers from search results.

Table A.1: Representative trajectories demonstrating our model’s learned search and reasoning capabilities. The model exhibits sophisticated behaviors including (a) recognition of irrelevant results, (b) query reformulation, and (c) verification strategies.

Example 1: Presidential Succession (<i>Query Refinement after Failure</i>)	
Question	Who is next in line if something happens to the president?
Gold	vice president
Turn 1	<code>search("who is next in line if something happens to the president")</code> → Retrieved: Documents about novels "Something Happened" and "Mosquitoes Buzz" Agent: "The queries were not directly about the line of succession... The relevant information I need is not present in the search results. Let’s try searching again with a more specific query."
Turn 2	<code>search("line of succession for the president")</code> → Retrieved: Philippine and US presidential succession information Agent: "Based on the search results... the next person in line would be the Vice President."
Answer	✓ The Vice President
Example 2: University Identification (<i>Information Verification</i>)	
Question	Which American University is situated in Ithaca, New York?
Gold	Cornell University
Turn 1	<code>search(["American University in Ithaca", "University in Ithaca NY"])</code> → Retrieved: "Ithaca is home to Cornell University, an Ivy League school..." Agent: "Ithaca, New York is home to Cornell University, which is an Ivy League school. However... it is highly likely that the university in question is Cornell. Let me confirm this."
Turn 2	<code>search(["Cornell University location", "Cornell University New York"])</code> → Retrieved: Cornell history confirming establishment in Ithaca Agent: "Cornell University is indeed located in Ithaca, New York."
Answer	✓ Cornell University
Example 3: Biographical Information (<i>Direct Extraction</i>)	
Question	What is John Andrew Martin’s occupation?
Gold	lawyer, attorney, politician
Turn 1	<code>search(["John Andrew Martin occupation", "John Andrew Martin job"])</code> → Retrieved: "American journalist, attorney, soldier, and politician..." Agent: "John Andrew Martin was an American <u>journalist, attorney, soldier, and politician</u>. He represented Colorado in the U.S. House of Representatives."
Answer	✓ Journalist, attorney, soldier, and politician